



Shreya Singh

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Professional Summary

Software Engineer with experience developing scalable backend systems, RESTful APIs, cloud-based applications, and AI-powered software solutions. Proficient in Java, Python, JavaScript, React.js, Flask, PostgreSQL, ANSI SQL, SDLC, Agile methodologies, and cloud technologies with hands-on experience building software applications and machine learning solutions.

Education

Bachelors of Technology - Computer Science and Engineering (Artificial Intelligence and Machine Learning) <i>Pranveer Singh Institute of Technology</i>	10/2023 – 06/2027 71.65%(till 6th semester)
Intermediate, U.P Kirana Seva Samiti Balika Vidyalaya	04/2021 – 06/2022 74.6%
High School, U.P Kirana Seva Samiti Balika Vidyalaya	04/2019 – 03/2020 85.6%

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript
Backend: Flask, REST APIs, Client-Server Architecture, CRUD Operations, API Lifecycle Management, Authentication & Authorization
Frontend: HTML5, CSS3, Responsive Web Design, Frontend-Backend Integration, React.js
AI/ML & Cloud: Microsoft Azure (Fundamentals), Google Cloud Vertex AI, Machine Learning, Generative AI, Data Preprocessing, Prompt Engineering, Scikit-learn, TensorFlow, Pandas, NumPy, Cloud Fundamentals
Database & Tools: ANSI SQL, PostgreSQL, Supabase, VS Code
Core Computer Science: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, SDLC, Agile/Scrum, Problem Solving, Debugging
DevOps & Deployment: Git, GitHub Actions, Docker, CI/CD Fundamentals, Deployment Pipelines, Render, Vercel

Experience

Explainable Skin Cancer Detection using CNN-Vision Transformer (ViT), Software Engineer (AI/ML) • Engineered a hybrid CNN-Vision Transformer (ViT) architecture using TensorFlow/PyTorch for skin lesion classification, improving accuracy through combined local and global feature extraction; implemented Explainable AI (XAI) using Grad-CAM and attention-map visualization, increasing model interpretability with human-readable diagnostic insights. • Developed Retrieval-Augmented Generation (RAG)-based LLM integration for medical reasoning and explainable output generation, improving clinical transparency and decision-support usability; built scalable ML pipelines using Python, Flask/Streamlit, NumPy, Pandas, and OpenCV for preprocessing, inference, and real-time HAM10000 image classification. • Collaborated in Agile/Scrum-based SDLC workflows within a 4-member team, improving development efficiency, experimentation cycles, and cross-functional coordination.	03/2026 – Present
AI-Powered Marketplace Assistant for Local Artisans, Full-Stack Software Developer • Engineered a cloud-native, microservices-based application using Python (Flask), JavaScript, RESTful APIs, and PostgreSQL, streamlining product onboarding through scalable backend services and API-driven workflows. • Implemented AI-powered content generation using Google Cloud Vertex AI and designed secure PostgreSQL database schemas to automate content generation, optimize query performance, and ensure efficient data management. • Applied Object-Oriented Programming (OOP) principles, Software Development Life Cycle (SDLC) practices, and Agile/Scrum methodologies to develop, test, and deploy features through collaborative and iterative development.	12/2025 – 02/2026
Road Accident Severity Prediction, Backend Software Developer • Developed a predictive machine learning model achieving 82% classification accuracy through data preprocessing, feature engineering, model training, and evaluation. • Built RESTful APIs using Python (Flask) to deploy real-time model inference and integrated backend services for scalable prediction workflows. • Optimized backend data pipelines for efficient processing of large datasets and applied Data Structures & Algorithms (DSA) concepts to improve algorithm efficiency and overall application performance.	08/2025 – 11/2025

Certifications

IBM Full Stack Software Developer Professional Certificate — Coursera
Machine Learning Specialization — Coursera

Extracurricular Activities

Flipkart GRID - 8.0 Qualified for Round 2 (Coding Round) of Flipkart GRiD, a national engineering campus challenge.	07/2026
E-Summit IIT Kanpur 2025, Attended Attended the E-Summit, gaining exposure to entrepreneurship, startup ecosystems, innovation strategies, and emerging technology trends through industry-led sessions and networking events	01/2025